

Question				Marking details	Marks Available					
					AO1	AO2	AO3	Total	Maths	Prac
5	(a)			Bohr effect / shift (1)	1			1		
	(b)	(i)		70 % (1) 10 % (1)		2		2	2	
		(ii)		Reduced affinity of haemoglobin for oxygen/ (oxy)haemoglobin dissociates at a higher partial pressure (of oxygen) (1) <u>More</u> oxygen {released / unloaded/ dissociates}/ oxygen released (more) readily (1) Allow oxygen more readily dissociated from haemoglobin = 2 marks	2			2		
	(c)	(i)		CO ₂ removed faster/ more CO ₂ exhaled (1) Increases {diffusion / concentration} gradient (from blood into alveoli) (1)			2	2		
		(ii)		increased affinity of haemoglobin for oxygen / decreased dissociation of oxyhaemoglobin (1) Less oxygen {released/ available} (for respiration) (1)			2	2		
	(d)	(i)		X = carbonic anhydrase Y = carbonic acid / H ₂ CO ₃ Z = hydrogen ion / H ⁺ / proton 0/1 = 0 marks 2 correct = 1 mark, 3 correct = 2 marks	2			2		
		(ii)		{In solution/dissolved} in plasma / as carbamino compounds / as carbaminohaemoglobin (1)	1			1		
		(iii)		Any three from: - Chloride shift (1) must be in context of red blood cells (chloride ions) {enter/diffuse} into red blood cells in exchange for hydrogen carbonate ions (1) - One to one exchange / reference to figures (1) - To maintain (electrochemical) neutrality (in red blood cell)/ to {equalise/ balance/ maintain} charge (1)	1	2		3		
				Question 5 total	7	4	4	15	2	0